Mobility protocol — lower body (hEDS / HSD safe)

Pre: joint control before warm-up sets · Post: joint check · active drills only (R7) · add-ons by target

Target area — dashed cards are add-ons

Glutes · Quads / knee · Hamstrings / adductors · Ankles / calves · Hips / balance / breath

⚠ Why active drills only — hEDS / HSD

Your joints have range to spare; what they lack is control. Muscles guard loose joints with protective tension, and passive stretching switches that guard off. Every drill here is made by your own muscles, inside the range they control, with nothing pushing further (R7).

PRE-WORKOUT — joint control (core ~11 min · before warm-up sets)

Core: every lower-body joint, every session · Add-ons: pick by your session's targets · light effort, slow breathing

Core — every session

P1 Ankle CARs, both feet together (seated) — 5 each direction
Ankle capsule · calves · shin muscles — ~10 s per circle
Seated, both feet lifted; both ankles circle slowly together (R12) — seated
The biggest circle you control, short of end range; note any catching (R1 · R7)

P2 Supine windshield wipers, both knees together — 5 each side
Hip rotators, in and out · hip capsule — slow, no bounce
On your back, knees bent, feet flat and wider than your hips
Both knees drop to one side to ~70–80% of the range, then the other; low back quiet (R1 · R3)

P3 Supine bent-knee leg lift to tabletop, both legs — 2 × 6 reps
Hip flexors · deep core — 2–3 s down
On your back, knees bent ~90°; lift both feet until the thighs are upright
Low back stays flat; lower slowly and stop where it starts to arch (R1 · R2)

P4 Supine banded butterfly (short band above the knees) — 2 × 10 reps
Glute med · hip abductors · side-of-hip control — short band
On your back, knees bent, feet flat hip-width; both knees open together
Open to ~70–80% of the range, not to the floor; pelvis and low back still (R1)

P5 Glute bridge, band above the knees, 3 s squeeze — 2 × 8 reps
Glute max · hamstrings — 3 s squeeze at the top
Knees pressed out against the band; ribs down, no arch in the low back (R3) — short band
⚠ The load peaks at the top: stop just short of full hip extension — that's your margin (R2)

P6 Short-band squat hold, knees pressed out (above parallel) — 2 × 20 s hold
Quads · glute med · knee tracking — short band
Short band just above the knees; press both knees out over the toes
Thighs stay above parallel at the deepest point; easy effort, no caving (R3 · R11)

P7 Calf-raise hold just short of the top, both legs — 2 × 8 reps
Calves · ankle control — 3 s hold each rep
Hand on the wall or rack; rise on both feet together (R12) — wall support
⚠ Hold just below your highest point: full height under body weight is end range (R2)

P8 Tandem stance, heel to toe (eyes open, then closed) — 2 × 30 s
Ankle and hip stabilisers · balance — one per front foot
Front heel against the back toes; knees soft; beside the rack, a hand ready
Eyes closed only once 30 s with eyes open is steady; swap the front foot (R23)

Add-ons — pick by your session's targets

P9 Banded knee extension, both legs (one band behind both knees) — 2 × 12 reps
Quads — before squat or knee-dominant work — light band
Band from the rack behind both knees; both knees work together (R12)
⚠ Straighten to just short of straight: the load peaks there, so never lock (R2)

P10 Hamstring hold, both heels pressing into the wall — 2 × 15 s hold
Hamstrings — before hinge work
On your back, both feet on the wall at ~45°, knees slightly bent (R2)
Press both heels into the wall ~30% and hold; both legs together (R12)

P11 Supine banded hip adduction, both legs (two long bands) — 2 × 10 reps
Adductors — before the hip machine or wide-stance work — 2 long bands
A long band round each knee from the frame, same pull both sides (R23) — frame
Start with the knees open to ~70–80% and bring both together slowly (R1)

POST-WORKOUT — joint check and wind-down (core ~3 min · after the last set)

Compare each joint with the pre half · log any change with the step gate (§2.6) · tightness → foam rolling (R7)

Core — every session

S1 Ankle CARs, slow scan, both feet together — 3 each direction
Joint check: compare with P1 — seated
Seated; 3 slow circles each way; note any new catching or stiffness
Worse than in P1 → note it in the step-gate log (§2.6)

S2 Supine windshield wipers, slow scan — 3 each side
Joint check: compare with P2
3 slow sweeps each side to ~70–80%; note any pinch, click or looseness
A loose or painful hip ends the drill; note it in the step-gate log (R9 · §2.6)

S3 90/90 breathing (feet on the bench or wall) — 5 breaths
Diaphragm · deep core — winding down — 4 s in / 8 s out
On your back, hips and knees at 90°, feet resting on the bench — ~1 min
Breathe in 4 s through the nose, ribs widen · out 8 s, ribs drop; easy, unforced

Add-ons — when your session calls for them

S4 Hamstring hold, both heels pressing into the wall — 2 × 15 s hold
Hamstrings — if they feel tight; foam rolling first (R7)
As P10: feet on the wall at ~45°, knees slightly bent (R2)
Press both heels ~30% and hold; both legs together (R12)

Holds: breathe in before, breathe out slowly during — never hold your breath
A slow exhale under tension keeps trunk pressure steady

Target index — what each drill prepares; pick add-ons by your session

Ankle · calves · Achilles	P1 · P7 · post S1
Hip rotation / capsule	P2 · post S2
Hip flexors / deep core	P3
Glute med / abductors	P4 · P6
Glute max	P5
Quads / knee tracking	P6 · add-on P9
Hamstrings	add-on P10 · post S4
Adductors	add-on P11
Balance	P8
Ribcage / winding down	post S3

Before vs after — log any change with the step gate (§2.6)

- Ankles: S1 vs P1 — any new catching or stiffness?
- Hips: S2 vs P2 — any new pinch, click or looseness?
- Knees: any pain, swelling or instability during or after the session?

Every drill, every session — hEDS / HSD rules

- Active only: your own muscles move and hold; nothing pushes a joint further, no bounce (R7)
- ~70–80% of your range, never end range; a soft bend at every joint, no lockout (R1 · R2)
- CARs: a spot catches or clicks → slow down there, never force through it (R7)
- Both legs together, or both feet loaded: one-leg drills wait for loading stage 3 (R12)
- Stage 1: knees above parallel; the low back moves only part-way, ribs down (R3 · R11)
- Light: holds at ~20–30% of a max push, moves slow and smooth — drills are not working sets (§2.3 · R19)
- A joint feels loose or unstable → stop that drill; loose joints that day → skip the session (R9)

Session timeline

Pre core (~11 min) → add-ons by target → 1–2 warm-up sets → your session → post core (~3 min)
Add-ons only when your session's targets call for them; tightness after a session → foam rolling (R7)

v3 — rebuilt from catalogue rev 13 at the current stages: core + add-ons by target, both legs together (R12), post half as a check